Simulation

□ 1. What is a “Good” Simulated Sensor?

A good simulation should:

✓ Produce **time-series data**
✓ Include **normal + abnormal behavior**
✓ Be **configurable**
✓ Mimic real-world patterns

□ 2. Three Recommended Approaches

✅Approach A — Random + Controlled Patterns (BEST START)

Students generate synthetic data with:

- Normal range
- Occasional anomalies
- Trend or periodic variation

Example: Temperature Sensor

```
import time
import random
import math

t = 0

while True:
    base_temp = 30 + 5 * math.sin(t/10)   # periodic variation
    noise = random.uniform(-1, 1)

    temp = base_temp + noise

    # Inject anomaly occasionally
```

```
    if random.random() < 0.05:
        temp += random.uniform(5, 10)

    print(temp)

    t += 1
    time.sleep(2)
```

Why this is powerful:

- Simulates real-world behavior (daily variation)
- Includes noise
- Includes anomalies → perfect for AI

✅Approach B — Dataset Replay (VERY USEFUL)

Students use real datasets and replay them as live streams.

Example:

```
import time
import pandas as pd

data = pd.read_csv("temperature_data.csv")

for _, row in data.iterrows():
    print(row["temperature"])
    time.sleep(2)
```

Where to get datasets:

- Kaggle
- UCI ML repository
- Public IoT datasets

Advantage:

- Realistic data
- Good for ML training

✅Approach C — Scenario-Based Simulation (ADVANCED)

Students simulate specific systems:

Examples:

System	Simulation
Motor	vibration increases before failure
Water tank	level rises and drops
Energy meter	peak usage at certain times
Room temp	HVAC cycles

Example: Water Tank

```
level = 50

while True:
  level += random.uniform(-2, 3)

  if level > 90:
     print("Overflow risk!")

  print(level)
  time.sleep(2)
```

□ 3. Standard Simulation Template

import time

```
import random
import json
import paho.mqtt.client as mqtt

BROKER = "YOUR_SERVER_IP"
TOPIC = "sensors/group01/project/data"

client = mqtt.Client()
client.connect(BROKER, 1883, 60)

t = 0

while True:
  value = 30 + 5 * random.uniform(-1, 1)

  # anomaly
  if random.random() < 0.05:
     value += 10

  data = {
     "value": round(value, 2),
     "timestamp": time.time()
  }

  client.publish(TOPIC, json.dumps(data))
  print(data)

  time.sleep(2)
```

□ 4. Types of Simulated Sensors You Should Encourage

□ Industrial

- Vibration sensor (sinusoidal + noise + spikes)
- Motor temperature
- Current / voltage

□ Environmental

- Temperature / humidity
- Air quality
- Water level

□ Signal-based

- Sound level (random + spikes)
- Event detection

□ 5. Making Simulation AI-Ready

Tell students:

□ “Your simulation must include anomalies”

Examples:

Type	Example
Spike	sudden jump
Drift	gradual increase
Noise	random variation
Pattern change	different behavior

□ 6. Difficulty Levels

□ Easy

- Random + threshold anomaly

□ Medium

- Periodic + noise + anomalies

 Advanced

- Multi-variable simulation (temp + humidity + energy)

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7. Common Mistakes to Avoid

✕Pure random data (no pattern)
✕No anomalies
✕Too fast data (overloads MQTT)
✕No timestamp

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8. What This Enables

With good simulation:

- AI models become meaningful
- Node-RED dashboards look realistic
- You can combine modules later into:
    - Industrial monitoring system
    - Commercial DAQ prototype

Final Recommendation

Give students:

✓ Standard simulation template
✓ Require:

- noise
- pattern
- anomalies

✓ Limit rate:

1 message every 2–5 seconds